

Deepak Mallampati

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PROFESSIONAL SUMMARY

Applied AI engineer (MS CS, 2.5+ years) specializing in GenAI-powered agentic workflows and multi-agent fraud risk systems for enterprise at scale. Built production multi-agent pipeline with schema-validated contracts, RAG-grounded validation, and ANN-based duplicate detection - directly applicable to adversarial risk investigation at billion-event scale. Delivered >30% hallucination reduction and 25% agent stability improvement across complex LLM workflows. Strong Python, LangChain, LlamaIndex, vector search, and client-facing enterprise delivery.

CORE COMPETENCIES

Multi-Agent Orchestration • GenAI Agentic Workflows • Fraud Risk Scoring Systems • RAG & Semantic Retrieval • Schema-Validated Agent Contracts • LangChain / LlamaIndex • Python / TypeScript • FastAPI / Docker • LLM Evaluation & Guardrails • Enterprise Client Deployment

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer - Healthcare Technology, AI Consulting

- Developed **GenAI-powered agentic workflows** for multi-step healthcare queries with structured reasoning, tool discipline, and grounding rules; improved agent stability by ~25% and reduced hallucinations by >30%.
- Built **RAG systems** with recursive semantic chunking and transformer embeddings; improved contextual retrieval precision by ~35%; retrieval-grounded alignment exceeded 90%.
- Shipped ML/LLM inference as **FastAPI/Flask RESTful services** with Docker; reduced post-deployment defects by ~30%.
- Built preprocessing pipelines for EHR extracts and support ticket logs; raised dataset reliability above 98%.
- Maintained HIPAA-conscious data governance: de-identification, data lineage, evaluation audit trails.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer

- RAG-grounded document Q&A; reduced hallucinations ~30%, raised accuracy >85%.
- Transformer-based video summarization; **60-70% reduction** in review time.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)

- SQL/T-SQL optimization; reduced query execution time ~20%, deployment errors >30%.

KEY PROJECTS

Agentic Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake → Validation → Consistency Analysis → Duplicate Detection → Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation; **ANN-based duplicate detection**; explainable risk scoring with audit-ready reasoning traces. Direct analog to billion-scale adversarial event classification.

Manga Lens - Multi-Provider AI Vision Integration

Production Chrome extension with four AI vision providers (Claude, GPT-4o, Ollama). Multi-provider payload abstraction - demonstrates cross-model inference architecture for high-throughput, reliability-first AI systems.

Driver Drowsiness Detection - YOLOv8 Real-Time Monitoring

Unified YOLOv8 detection; ~30% **latency reduction**, ~25% **false positive reduction** via sliding-window confidence aggregation.

EDUCATION

Master of Science, Computer Science - Kent State University, Ohio

Aug 2023 - May 2025

SKILLS

Languages: Python, TypeScript, JavaScript, SQL

LLM & GenAI: RAG, agentic workflows, LangChain, LlamaIndex, prompt engineering, vector search, structured outputs, guardrails

ML: PyTorch, scikit-learn, XGBoost, HuggingFace, YOLOv8

Infra: Docker, FastAPI, Flask, Jenkins, Grafana, REST APIs

Work Authorization: F-1 OPT (US-authorized now; future sponsorship required)